

Confidence Intervals for the Population Mean

Statistics Worksheet · Grade 11–12 / AP Statistics

Name: _____

Date: _____

Learning Objectives

- Identify and organize given values (n , $\bar{x}$, σ , confidence level) from a word problem
- Verify the two conditions required for a valid confidence interval (random sample, normal population or large n)
- Calculate the margin of error and confidence interval using the formula $\bar{x} \pm z^* \cdot (\sigma / \sqrt{n})$

Problems

1. A simple random sample of 100 students has a mean SAT math score of 520. The population standard deviation is known to be 80. Identify and list all four given values needed to construct a confidence interval.

n , $\bar{x}$, σ , confidence level

2. State the two conditions that must be verified before constructing a confidence interval for a population mean, and explain why each matters.

3. Find the critical value z^* for a 90% confidence interval using the formula shown. Then find z^* for a 99% confidence interval.

$$z^* \text{ where } \alpha/2 = \frac{1 - C}{2}$$

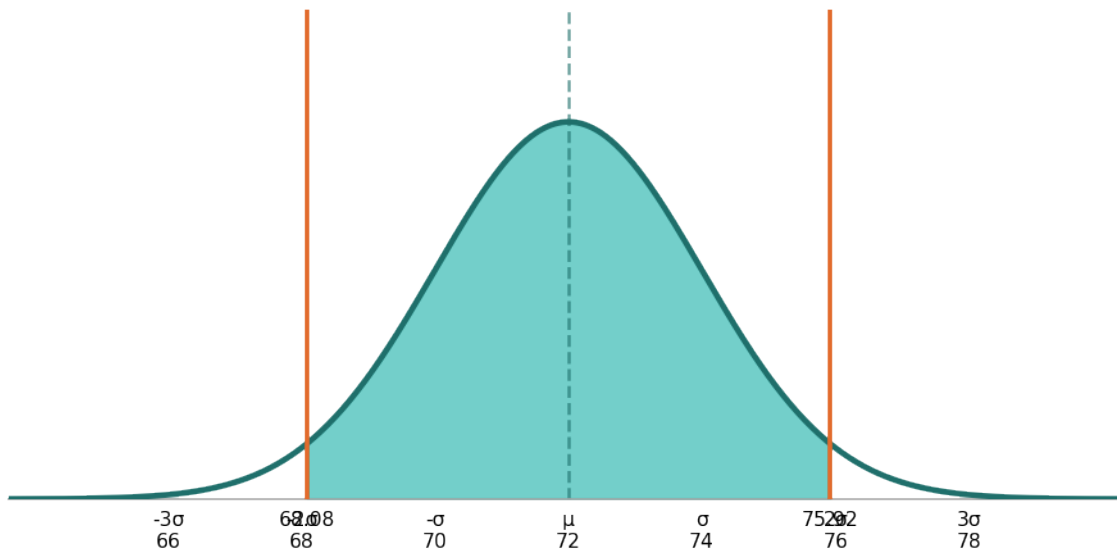
4. From a normally distributed population, a simple random sample of 64 observations has a mean of 72. The population standard deviation is 16. Calculate the standard error of the mean.

$$SE = \frac{\sigma}{\sqrt{n}}$$

5. Using the sample in Problem 4 ($n = 64$, $\bar{x} = 72$, $\sigma = 16$), construct the 95% confidence interval for the population mean. Use $z^* = 1.96$.



95% Confidence Interval ($\mu = ?$, $\bar{x} = 72$)

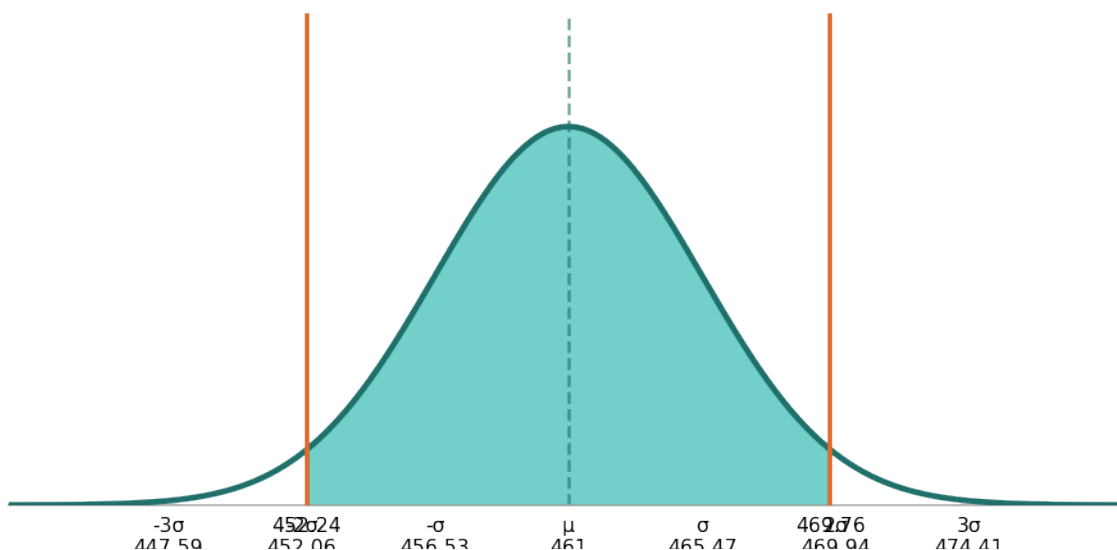


6. A researcher takes an SRS of 500 students and finds a mean SAT math score of 461. The population standard deviation is 100. Calculate the margin of error for a 95% confidence interval. Use $z^* = 1.96$.

$$ME = z^* \cdot \frac{\sigma}{\sqrt{n}}$$

7. Using the data from Problem 6 ($n = 500$, $\bar{x} = 461$, $\sigma = 100$), construct the full 95% confidence interval and write a sentence interpreting the result in context.

95% CI for Mean SAT Math Score



8. A normally distributed population has a standard deviation of 25. How large a simple random sample is needed so that the margin of error for a 95% confidence interval is no more than 5 points? Use $z^* = 1.96$.

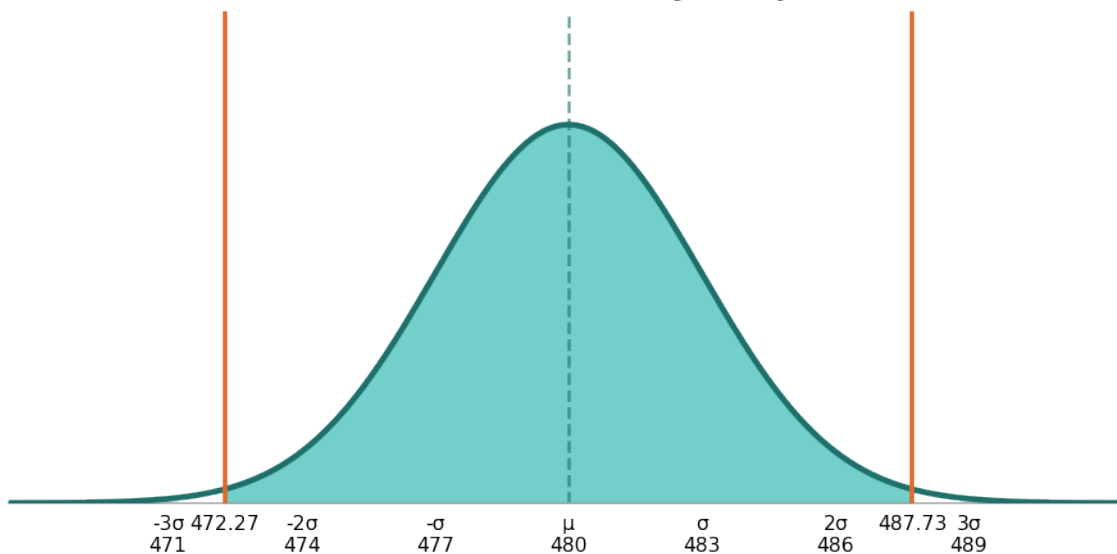
$$n \geq \left(\frac{z^* \cdot \sigma}{ME} \right)^2$$

9. Two students each take a simple random sample from the same normally distributed population ($\sigma = 50$). Student A uses $n = 100$ and gets $\bar{x} = 310$. Student B uses $n = 400$ and gets $\bar{x} = 310$. Both construct 95% confidence intervals. Find both intervals and explain which is narrower and why.

$$\bar{x} \pm 1.96 \cdot \frac{50}{\sqrt{n}}$$

10. A quality-control engineer takes an SRS of 225 batteries from a normally distributed production line and records a mean lifespan of 480 hours. The population standard deviation is 45 hours. Construct the 99% confidence interval for the true mean lifespan, verify both conditions, and write a conclusion statement. Use $z^* = 2.576$.

99% CI for Mean Battery Lifespan



Confidence Intervals for the Population Mean — Answer Key

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Answer Key

1. Answer: $n = 100$, $\bar{x} = 520$, $\sigma = 80$, $C = 0.95$ (assumed standard)

- Step 1: Sample size $n = 100$ (from 'SRS of 100 students').
- Step 2: Sample mean $\bar{x} = 520$ (from the SRS).
- Step 3: Population standard deviation $\sigma = 80$ (given).
- Step 4: Confidence level $C = 0.95$ (stated or assumed).

2. Answer: 1) Sample is randomly selected. 2) Population is normally distributed (or $n \geq 30$ by CLT).

- Condition 1: The sample must be randomly selected so results are unbiased and representative of the population.
- Condition 2: The population must be normally distributed (or the sample size must be large enough for the Central Limit Theorem to apply) so the sampling distribution of $\bar{x}$ is normal.

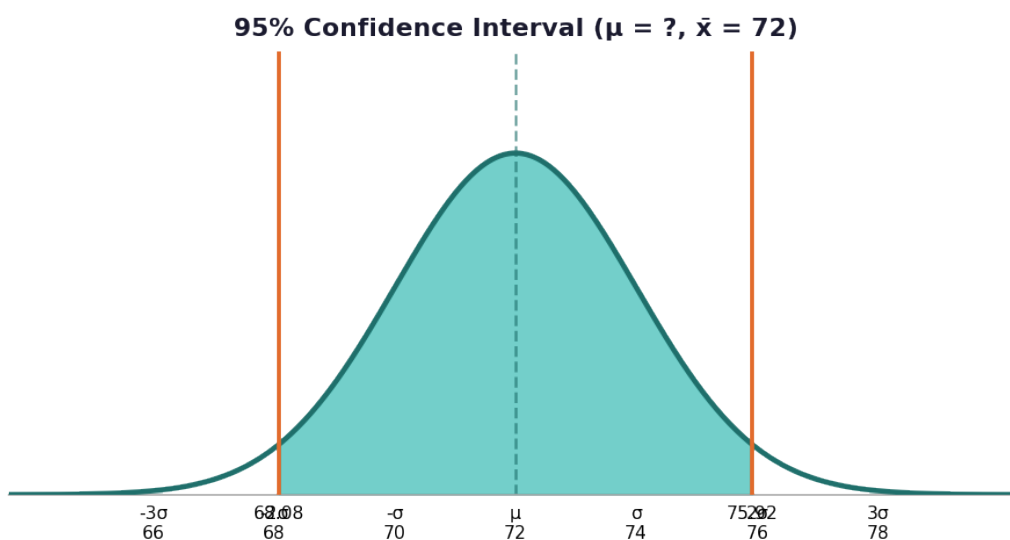
3. Answer: z^* for 90% = 1.645; z^* for 99% = 2.576

- For 90%: $\alpha/2 = (1 - 0.90)/2 = 0.05$. Look up z with area 0.95 in the z -table $\rightarrow z^* = 1.645$.
- For 99%: $\alpha/2 = (1 - 0.99)/2 = 0.005$. Look up z with area 0.995 in the z -table $\rightarrow z^* = 2.576$.

4. Answer: $SE = 2$

- Step 1: Identify $\sigma = 16$ and $n = 64$.
- Step 2: $SE = \sigma / \sqrt{n} = 16 / \sqrt{64} = 16 / 8 = 2$.

5. Answer: 95% CI: (68.08, 75.92)



- Step 1: $SE = 16 / \sqrt{64} = 2$.

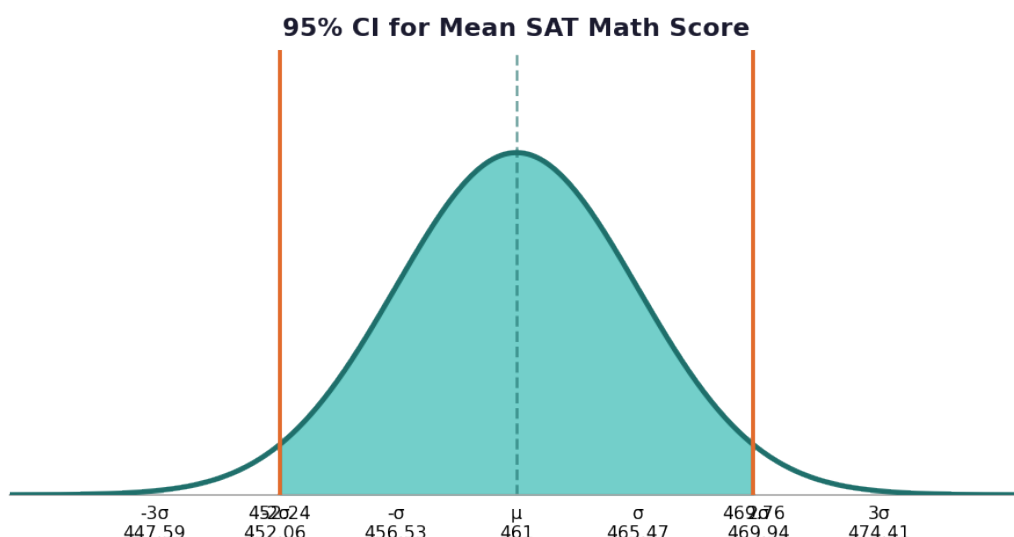


- Step 2: Margin of error = $z^* \times SE = 1.96 \times 2 = 3.92$.
- Step 3: Lower bound = $72 - 3.92 = 68.08$.
- Step 4: Upper bound = $72 + 3.92 = 75.92$.
- Step 5: 95% CI: (68.08, 75.92).

6. Answer: ME $\approx$ 8.76

- Step 1: $SE = \sigma / \sqrt{n} = 100 / \sqrt{500} \approx 100 / 22.36 \approx 4.47$.
- Step 2: $ME = z^* \times SE = 1.96 \times 4.47 \approx 8.76$.

7. Answer: 95% CI: (452.24, 469.76). We are 95% confident the true mean SAT math score lies between 452.24 and 469.76.



- Step 1: $ME = 1.96 \times (100 / \sqrt{500}) \approx 1.96 \times 4.47 \approx 8.76$.
- Step 2: Lower bound = $461 - 8.76 = 452.24$.
- Step 3: Upper bound = $461 + 8.76 = 469.76$.
- Step 4: Interpretation: We are 95% confident the true population mean SAT math score lies between 452.24 and 469.76 points.

8. Answer: $n \geq 97$ (round up to 97)

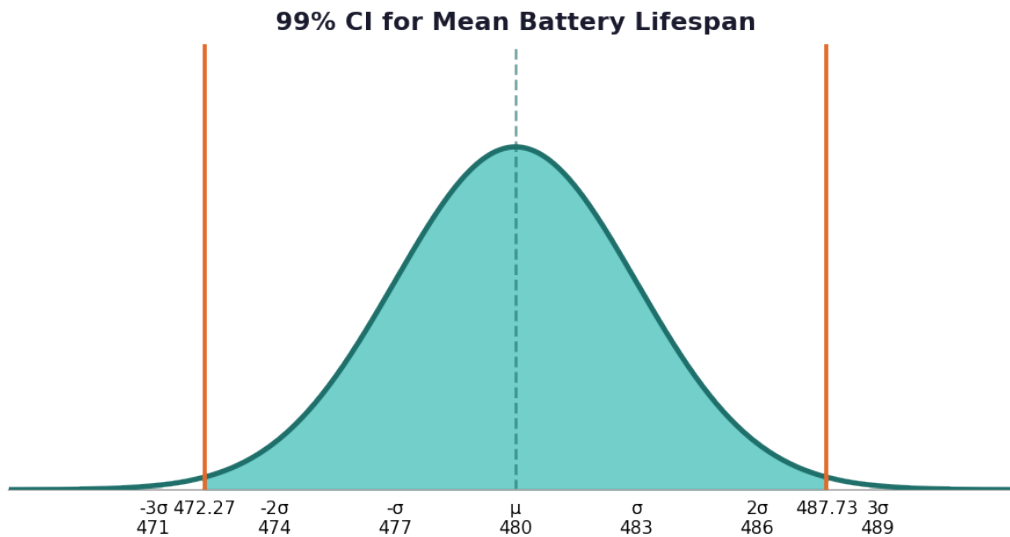
- Step 1: Set up the inequality: $ME \leq 5$, so $z^* \cdot \sigma / \sqrt{n} \leq 5$.
- Step 2: Solve for n : $\sqrt{n} \geq (1.96 \times 25) / 5 = 49 / 5 = 9.8$.
- Step 3: $n \geq 9.8^2 = 96.04$.
- Step 4: Round up to the nearest whole number: $n \geq 97$.

9. Answer: Student A: (300.2, 319.8); Student B: (305.1, 314.9). Student B's interval is narrower because larger n reduces SE.

- Student A: $SE = 50 / \sqrt{100} = 5$; $ME = 1.96 \times 5 = 9.8$; $CI = (310 - 9.8, 310 + 9.8) = (300.2, 319.8)$.
- Student B: $SE = 50 / \sqrt{400} = 2.5$; $ME = 1.96 \times 2.5 = 4.9$; $CI = (310 - 4.9, 310 + 4.9) = (305.1, 314.9)$.
- Student B's interval is narrower because a larger sample size reduces the standard error, which reduces the margin of error.
- Both intervals are centered at $\mu = 310$, but Student B's width is only 9.8 vs. Student A's 19.6.



10. Answer: 99% CI: (472.27, 487.73). We are 99% confident the true mean battery lifespan is between 472.27 and 487.73 hours.



- Condition 1: The sample is a simple random sample — satisfied.
- Condition 2: The population is normally distributed — satisfied (stated in problem).
- Step 1: $SE = \sigma / \sqrt{n} = 45 / \sqrt{225} = 45 / 15 = 3$.
- Step 2: $ME = z^* \times SE = 2.576 \times 3 = 7.728 \approx 7.73$.
- Step 3: Lower bound = $480 - 7.73 = 472.27$.
- Step 4: Upper bound = $480 + 7.73 = 487.73$.
- Conclusion: We are 99% confident the true mean battery lifespan lies between 472.27 hours and 487.73 hours.

